VRUKSHAAYURVEDA

OF VARAAHAMIHIRA

Dr. N. Gopalakrishnan

Indian Institute of Scientific Heritage
Thiruvananthapuram

Heritage Publication Series - 17

Scanned with OKEN Scanner

VRUKSHAAYURVEDA

OF VARAAHAMIHIRA

Dr. N. Gopalakrishnan

Indian Institute of Scientific Heritage

Thiruvananthapuram

Heritage Publication Series - 17

Scanned with OKEN Scanner

VRUKSHAAYURVEDA of Varahamihira

Dr. N. Gopalakrishnan M.Sc. (pharm), M.Sc. (Chem), M.A. (Soc), Ph.D
(Scientist, CSIR)

Published by :

Indian Institute of Scientific Heritage (IISH)
Ushus, Estate Road, Pappanamcode
Trivandrum - 695 018 (Ph. 490149)

Rs. 10/-

Printed at:

Sree Printers (DTP, Offset & Screenprinting)
Ind. Estate, Pappanamcode, TVM - 19, Ph. 490135

Publication Conveners

Sri. K.V. Unnikrishnan, Ph: 0471 - 460707
Smt. K. Subhadramma, Ph: 0471 - 347486
(RRL, CSIR, Trivandrum-19)

Scanned with OKEN Scanner

Dhanyathman,

Ayurveda is the ancient Indian scientific subject dealing with the health and age of the individuals. It is the Upaveda of Atharva veda. It deals with diseases, their cause, solution, treatment and procedures, for keeping good health. Susrutha Samhitha and Charaka Samhitha are the two important literatures in the Ayurvedic subject dealing with the health of human beings. Ashtangahrudaya and Ashtangasamgraha are also equally important books of later origin. There are many books written in Sanskrit and Regional languages on the health of animals. Books like Hasthyaayurveda, Gajaayurveda, Aswaayurveda discuss various aspects on the health of elephants and horses. Treatment methods given in these books are also useful for other animals.

Like the health aspects of animals and people, the treatment and health of the plants were also looked into in detail in India. Variety of measures are advocated for maintaining good health for the plants, and keeping high yield from them. Agricultural practices, use of pesticides, particularly bio-pesticides, hormonal types of treatment using peculiar fermented products, etc. were also the subject of the Vrukshaayurveda texts written two to three thousand years ago. Some of them were also written before 500 AD. The plant science and agricultural aspects have been discussed in detail by Sarngadhara in his famous Vrukshaayurveda. Bhishmachaarya in Mahabharath has given the life forces in the plants. Charaka samhitha gives the detailed classification of the plants. They have also gone deep into the effectiveness of the medicinal plants depending upon the place of their collection/ cultivation. Indirectly this is a phytochemistry subject.

The important point worth remembering here is the approach

Scanned with OKEN Scanner

followed by ancient Indians on the treatment of the plants, thousands of years ago, which was similar to the procedure followed in the treatment of human beings and animals. Whereas the modern West could understand that the plants are living beings, only in the beginning of the 20th century AD, after Acharya Jagadeesh Chandra Bose's scientific explorations. It is also noteworthy that the agricultural practices including the grafting method suggested by Varahamihira and Kashyapa in their respective Vrukshaayurveda books were similar in approach to the modern techniques. Thus the Indian approach in the plant science was as excellent, years ago as in modern science in all the basic approach. Hence it is the bound privillege of Indian Institute of Scientific Heritage to spread the message of this branch of ancient plant science.

Here the 55th chapter of the Bruhath samhitha of Varahamihira (505 AD) is explained and this chapter is known as the Vrukshayurveda of Varahamihiraa. Wherever possible the modern scientific information are also quoted for comparison.

Indian Institute of Scientific Heritage is submitting this book to the well wishers of the Institute for learning and teaching, expecting your blessings and cooperation for the patriotic mission undertaken by the Institute, for spreading the scientific messages of this land.

Dr. M. Sambasivan
Chairman

Dr. N. Gopalakrishnan
Hon. Director

Scanned with OKEN Scanner

VRUKSHAAYURVEDA

It is well known from the history book that Emperor Asoka took initiative to plant trees on the road sides to provide shades for the pedestrians and animals. However there was also a concept in ancient India on planting the trees for aesthetic beauty and use for entertainment parks. This is very clear from the first line of the Vrukshayurveda written by Varahamihira.

प्रान्तच्छायाविनिर्मुक्ता न मनोज्ञा जलाशयाः।
यस्मादतो जलप्रान्तेष्वारामान् विनिवेशयेत्।।

Praanthacchaayaavinirmukthaa na manojnaa jalaasayaa:
yasmadatho jalaprantheshvaaraamaan vinivesayeth

Since the unshaded banks of lakes and reservoirs are not very attractive, parks and gardens should be established in these places.

The concept of parks was existing thousands of years ago in India and it is the same as the modern concept included in the town planning.

मृद्वी भूः सर्ववृक्षाणां हिता तस्यां तिलान् वपेत्।
पुष्पितान् तांश्च गृह्नीयात् कर्मैतत् प्रथमं भुवः।।

Mrudvee bhoo: sarvavrukshaanaam hithaa thasyaam thilaan vapeth
pushpithaan thaamscha gruhneeyaath karmaithath prathamam bhuvee:

Since a soft and mellow soil is beneficial to all types of trees, sesame (gingelly) seeds should be sown in places where the trees are to be planted, when these gingelly sprout and come into flower, they should be transplanted or incorporated with the soil. This treatment is to be rended to soils.

Growing trees in hard laterite type of soils is very difficult. Hence the leaves and stems of sesame which contain large amount

Scanned with OKEN Scanner

of mucilage, that will be helpful to give a soft texture for the so and thus air and water retention capability be increased. Th method is highly scientific.

अरिष्टाशोकपुन्नागशिरीषाः सप्रियङ्गवः। मङ्गल्याः पूर्वमारामे रोपणीयाः गृहेषु

Arishtaasokapunnaagasireeshaa: sapriyangava: mangalyaa: poorvamaraame ropaneeyaa: gruheshu vaa

Arishta (the neem tree) Asoka (Saraca indica) punnag (Cobra saffron tree) sirisha (Acacia speciosa) with the Priyang creeper are suitable for initial planting in parks and dwellings.

Here the variety of plants with medicinal values and f aesthetic beauty are suggested to be grown in the parks.

पनसाशोककदलीजम्बूलकुचदाडिमाः द्राक्षापालीवताश्चैव
बीजपूरातिमुक्तकाः ऐते द्रुमाः काण्डरोप्याः गोमयेन प्रलेपिताः।
मूलोच्छेदेfथवा स्कन्धे रोपणीयाः परं ततः

Panasaasokakadaleejamboolakuchadaadimaa: draakshaapaaleevathaaschaiva beejapooraathimukthakaa: eth drumaa: kaandaropya: gomayena pralepithaa: moolocchedeathavaa skandhe ropaneeyaa: param thatha:

The Indian jack tree, Euginia jambos, Artocarpus lakooch pomegranate, the grapes, Cassia cinnamon, the citron, Dalberig ougeinensis are to be planted.

The fruit yielding plants and spices are also included in th list, hence the space utilisation and economical aspects are balance in this approach. Thus an economical outlook and usefulless plu the aesteic beauty were integrated in the ancient park making.

अजातशाखान् शिशिरे जातशाखान् हिमागमे।
वर्षागमे च सुस्कन्धन् यथा दिकस्थान् प्ररोपयेत्।।
Ajaathasaakhaan sisire jaathasaakhaan himaagame

Scanned with OKEN Scanner

varshaagame cha suskandhaan yathaa diksthaan praropayeth

Clones without branching should be grafted in autumn, sprouted branch should be grafted in spring (hemantha), well developed branches should be grafted in the rainy season, by looking into selection of the direction of the grafting one should graft the branches.

Different types of twigs are used for grafting on the host plants. The Scientific guidance given here is based on the season. It is true that, if twigs required for grafting is not properly selected, they may dry off depending upon the ambient temperature and water availability. Hence the scientific outlook on the environmental conditions, as suggested in the modern method, is presented here.

घृतोशीरतिलक्षौद्रविडङ्गक्षीरगोमयैः। अमूलस्कन्धलिप्तानां सङ्क्रामणविरोपणम्।।

Ghruthoseerathilakshoudravidangaksheeragomayai:
amoolaskandhalipthaanaam sankraamanaviropanam

The clones should be smeared all over with a medication of ghee, Andropogon muricatus plant, sesamum, Embelia ribes, milk and cow dung before their transgrafting or transplanting.

Recent studies have shown that cow dung is an excellent source of useful micro organisms which can protect the plants from pathogenic organisms. Coating of oil or ghee (of course ghee is costly) can remove all the harmful microbes and the milk provide rich nitrogen source of protein, water and salts for growing the useful micro organisms present in the cowdung, which may protect the twigs/plants during the process of grafting or transplanting.

शुचिर्भूत्वा तरोः पूजां कृत्वा स्नानानुलेपनैः
रोपयेद्रपिताश्चैव पत्रैस्तैरेव जायते

Suchirbhoothwa tharo: poojaam kruthwa snaanaanulepani:
ropayedropithaschaiva pathraisthaireva jaayathe

Scanned with OKEN Scanner

Having cleaned oneself, (after taking bath) the tree should be worshipped and bathed after which the medicated pastes may be applied. If planted in this manner the clone will flourish without any set back.

Modern science may have nothing to say on the prayer before planting. However for the last one decade, the approach on the environment has been changed and a holistic outlook to air, water, plants, trees, earth, etc., is spreading. These lines may have validity only under above outlook.

सायं प्रातश्च घर्मर्तौ शीतकाले दिनान्तरे।
वर्षासु च भुवः शोषे सेतव्या रोपिता द्रुमाः
Saayam praathascha gharmanthe seethakaale dinaanthare
varshaasu cha bhuva: soshe sethavyaa ropithaa drumaa:

During summer, the newly established plants should be irrigated in the morning and evening, during winter on alternate days and during rains, whenever the grounds dry up.

जम्बूवेतसवानीरकदम्बोदुम्बरार्जुनाः वीजपूरकमृद्वीकालकुचाश्च
सदाडिमाः वञ्जुलो नक्तमालाश्च तिलकाः पनसास्तथा
तिमिरोऽम्रातकश्चेति षोडशानूपजाः स्मृताः
Jamboovethasavaaneerakadambodumbaraarjunaa:
beejapoorakamrudveekaalakuchaascha sadaaddimaa:
vanjulo nakthamaalaascha thilakaa: panasasthathaa
thimiroamraathakaschethi shodasaanoopajaa: smruthaa:

Terminelia arjuna, the citron, Vinifera grape, pomegranate, dalbergia, Pongamia glabra, wild cinnamon, jack fruit, ash pumpkin, Indian hog plum, are suitable for low lying well irrigated lands.

These lines give good example on the experienced scientific outlook of the ancient Indians on the agricultural practices. This

Scanned with OKEN Scanner

must definitely be based on their direct observations and experiments in the farming. It cannot be based on the so called intuition of the Indians.

उत्तमं विंशतिर्हस्ता मध्यमं षोडशान्तरम्।
स्थानात् स्थानान्तरं कार्यं वृक्षाणां द्वादशावरम्
Utthamam vimsathirhasthaa madhyamam shodasaantharam
sthaanath sthaanantharam kaaryam vrukshaanaam dwaadasaavaram

The optimum distance for planting is 20 cubit (which is equal to 30 feet) medium distance is 16 cubit and the minimum spacing should be12 cubit and further closer is not desirable.

Approximately ten metres distance (~33 ft) is the ideal space suggested for all trees by the modern science, which is also well in agreement with the ancient approach. Whenever space problems exist, reduction in this value is given, as also agreed by modern science. Interestingly the reason given is the exact science.

अभ्यासजातास्तरव: संस्पृशन्त: परस्परम्।
मिश्रैर्मूलैश्च न फलं सम्यग्यच्छन्ति पीडिता:
Abhyasajaathaasthaarava: samsprusantha: parasparam
mishrairmoolaischa na phalam samyagyacchanthi peedithaa:

Trees which are planted so close, that their canopies touch, will not yield fruit since they cause them to be mutually tortured on account of their root and crown systems intertwining.

Due to the reduced sunlight and also lack of availability of nutrients and water the leaves and also the roots may not function properly. Hence the growth and yield of the trees/fruits will not be proper, which is clearly explained by Varahamihira.

शीतावातातपै रोगो जायते पाण्डुपत्रता।
अवृद्धिश्च प्रवालानां शाखाशोषो रसश्रुति:
Seethavaathaathapai rogo jaayathe paandupathrathaa

Scanned with OKEN Scanner

avruddhischa pravaalaanaam saakhaasosho rasasruthi:

Cold air disturbances due to climatic change, excess heat o sunshine etc. cause diseases which may be recognised by the pallo in the foliage, its scaliness, sparsity or absence of shoots, wiltin or drying up of the branches or bleeding sores in any part of the tree.

Frost and hot weather can cause problems on the plants, which will be resulted in the major growth problems in the course of continuous exposure to these conditions. Above observations are also true according to modern science.

चिकित्सितमथैतेषो शस्त्रेणादौ विशोधनम्।
विडङ्गघृतपङ्काक्तान् सेचयेत् क्षीरवारिणा।

Chikitsithamathaitheshaam sasthrenaadow visodhanam
vidangaghruthapankaakthaan sechayeth ksheeravarinaa

Treatment of the affected trees can be accomplished by excising the diseased portion with suitable tool and applying a paste of vidanga (Ebelia ribes) ghee and mud and spraying the whole plant with a mixture of milk and water which will result in curing the diseases.

Cutting off the branches which are affected with the infectious diseases is the scientific practice. For avoiding further spreading of the pathogenic organisms, mud smearing is advocated so that the beneficial bacteria present in the mud can destroy the pathogenic organisms and also the presence of the plant extract of vidanga can have some medicinal properties to destroy the microorganisms in the new plant. However if care is not taken in this matter some other pathogenic organisms present in the mud can cause injury to the plant. Spray of ghee or milk etc. can prevent the growth of pathogenic organisms by way of forming a layer of fat on the affected surface of the plant.

Scanned with OKEN Scanner

फलनाशे कुलत्थैश्च माषैर्मुद्गैस्तिलैर्यवैः।
श्रृतशीतपयस्सेकः फलपुष्पसमृद्धये।।

Phalanaase kulaatthaischa maashairmudgaisthilairyavai:
sruthaseethapayasseka: phalapushpasamruddhaye

अविकाजशकृच्चूर्णस्याढके द्वे तैलाढके।
सक्तुप्रस्थो जलद्रोणो गोमांसतुलया सह।।

Avikaajasakruchhoornasyaadake dve thailadake
sakthuprastho jaladrono gomaamsathulaya saha

सप्तरात्रोषितैरेतैः सेकः कार्यो वनस्पतेः
वल्लीगुल्मलतानां च फलपुष्पाय सर्वदा

Saptharaathroshithairethai: seka: kaaryo vanaspathe:
Valleegulmalathaanaam cha phalapushpaaya sarvadaa

One adaka each of the powdered voiding of goat and sheep are taken together with powdered sesamum seed, fried barely, grist in the amount of two adakas and one prastha respectively in one drona of water placing in it one tula of meat. This mixture should be allowed to ferment over seven nights. If this fermented broth is fed to the trees and plants they will be bearing flowers and fruits continuously.

No practical study results on this aspect has been reported. And it may be necessary to conduct research work on this, rationally and with open mind, before concluding any aspect of the above explanation. However the above ferment is found to be an ideal source of nitrogenous material which will be directly absorbed through roots, and hence advantageous.

वासराणि दश दुग्धभावितं बीजमाज्ययुतहस्तभावितम्
गोमयेन बहुशो विरूक्षितं। क्रौडमार्गपिशितैश्च धूपितम्।।

Scanned with OKEN Scanner

Vaasaraani dasa dugdhabhaavitham
beejamaajyayuthahasthabhaavitham gomayena bahuso
virookshitham kroudamaargapisithaischa dhoopitham

The seeds must be soaked daily in milk, handling only afte smearing (the hands before touching the seeds) with ghee for te days, after which they must be coated with cow dung paste an finally fumigating them with smoke from burning pieces of dee flesh and boar flesh after placing them in a pot. The fumigatio technique is well known in the modern science. Role of the fatt material may only be for protecting the seeds from pathogeni microorganisms.

मांससूकरवसासमन्वितं रोपितं च परिकर्मितावनौ।
क्षीरसंयुतजलावसेचित। जायते कुसुमयुक्तमेव तत्।।
Maamsasookaravasaasamanvitham ropitham cha
parikarmithaavanow ksheera samyuthajalaavasechitham
jaayathe kusumayukthameva thath

The seeds should be sown in earth that has been mixed with flesh trimings and boar fat and treated by sowing sesamum seeds as explained in the verses 3. Those seeds were watered with a mixture of milk and water. When they germinate and start bearing flowers and fruits, hearly.

तिंत्रिनीत्मपि करोति वल्लरीं व्रीहिमाषतिलचूर्णसक्तुभि:
पूतिमांससहितैश्च सेचिता धूपिता च सततं हरिद्रया।।
Thinthrineemapi karothi vallareem vreehimaashathilachoornasakthubhi:
poothimaamsasahithaischa sechithaa dhoopithaa
cha sathatham haridrayaa

Even tamarind can be sprouted if a mixture of rice, sesame, black gram, barley, grist and staele carrion meat be used and continuously fumigated with burning pieces of turmeric root. Hence it can also be suggested for many other seeds.

Scanned with OKEN Scanner

कपित्थवल्ल्लीकरणाय मूलान्यास्फोतधात्रीधववासिकानाम्
पलाशिनी वेतससूर्यवल्ली स्यामातिमुक्तैः सहिताष्टमूली।।

Kapitthavaleekaranaaya
moolaanyaasphothadhaathreedhavavaasikaanaam palaasinee
vethasasooryavallee syaamaathimukthai: sahithaashtamoolee

The principle of germinating kapittha seeds is as follows. A medicinal decoction is prepared by boiling in milk, roots of Calotropsis gigantia, Emblica officinalis, Woodforde floribunda and Adathoda vasika, the plants Calamus rotang, wild dog mustard, and Ichnocarpus frutescence, Hemedesmus anatamul, Jasminum pubens then cooled before use.

Majority of the above mentioned products are well known medicinal plants used in the treatment, through ayurvedic preparations.

क्षीरे श्रृते चाप्यनया सुसीते तालाशतं स्थप्य कपित्थबीजम्।
दिने दिने शोषितमर्कपादैर्मासं विधिस्तवेष ततोऽधिरोप्यम्।।

Ksheere srithe chaapyanayaa suseethe thaalaasatham sthaapya
kapitthabeejam dine dine soshithamarkapaadairmaamsam
vidhisthavesha thathoadhiropyam

The ferronia seeds should be placed in the above medicated liquor daily for a duration of 100 seconds and dried in the Sun, repeating the process daily for one month and the seeds will be ready for sowing (the seeds will germinate fast)

It is an interesting subject for doing research. Can there be any plant hormonal type of activity in the above decoction?

हस्तायतं तद्विगुणं गभीरं खात्वावटं प्रोक्तजलावपूर्णम्।
शुष्कं प्रदग्घ मधुसर्पिषा तत् प्रलेपयेद् भस्मसमन्वितानाम्

Hasthaayatham thadvigunam gabheeram khaathvaavatam
prokthajalaavapoornam sushkam pradagdham madhusarpishaa

Scanned with OKEN Scanner

thath pralepayeth bhasmasamanvithaanaam

A pit should be dug one cubit wide and two cubit deep. Fille with a mixture of milk and water and allowed to dry out after whic a fire should be set with some dry materials thoroughly, after whic that pit should be smeared on the inside with a mixture of honey ghee and wood ashes.

The firing destroys all the (pathogenic) organisms presen in the pit and gives the germinating seeds a hygenic surroundings The ash produced by burning all these, gives a source of potassiun and other micronutrient mineral for the plant growth.

चूर्णीकृतैर्माषतिलैर्यवैच प्रपूरयेन्मृत्तिकयान्तरस्तै। :।
मत्स्यामिषाम्भस्सहितं च हन्याद्यावद्धनत्वं समुपागतं तत्

Choorneekruthairmaashathilairyavaischa
prapoorayenmrutthikayaantharasthai: mathsyaamishambhussahitham
cha hanyaadyaavaddhvanathvam samupaagatham thath

Thereafter the mixture of powdered black gram, green gram sesamum and barely should be obtained and the pit filled up with soil and the lintil grain powder in the following manner. Three inches thick layer of soil (4 angulas) is put in the pit and lintil grain mixture spread on it consolidating it by beating sprinkling with water containing pieces of flesh and fish. The process is repeated until the entire pit is filled up. The earth during filling is beaten hard, sprinkling it with cold aqueous infusion of fish and flesh.

By this method an extremely rich source of organic manure, micronutrient, soft surrounding for the seed to sprout, and highly moist seed bed etc. will be made available to fasten the seed germination and maintaining healthy seedlings.

The peas and grams are well known sources of nitrating and nitrifying bacteria in their roots and hence they enriches the soil

Scanned with OKEN Scanner

with nitrogenous materials. Thus the yield of the plant will definitely increse as noticed in the modern agricultural practices.

उप्तं च बीजं चतुरङ्गुलाधो मत्स्याम्भसा मांसजलैश्च सिक्तम्।
वल्लीभवत्याशु शुभप्रवाला विस्मापनी मण्डपमावृणोति।।
Uptham cha beejam chathurangulaadho mathsyambhasaa maamsajalaischa siktham valleebhavathyaasu subhapravaalaa vismaapanee mandapamaavrunothi

The treated seeds are sown at a path of 4 angulas (3 inch) after loosening the soil as required to the proper depth and watered with the meat and fish infusion. Then the seeds will readily germinate covering the entire basin with vigorous shoots in the most astonishing manner.

The seeding procedure is in agreement with the modern practice. The best source of nitrogen is already made available for the newly germinating plant so the plant will grow healthy and fast.

शतशोङ्गोलसम्भूतफलकल्केन भावितम् एततैलेन वा बीजं श्लैष्मातकफलेन वा
Sathasoangolasambhoothaphalakalkena bhaavitham ethathailena vaa beejam slaishmaathakaphalena vaa

Seeds are treated a hundred times with juice or pulp of angola fruits (Alanguim hexapetalum = sage leaved alanguim) or in the juice of shleshmaataka (Cordia latifolia - Sebastian plum).

Some studies have been carried out on the use of angola extract and some beneficial role in the seed sprouting could be noted. However indepth study is to be carried out for getting better results out of this ancient knowledge.

वापितं करकोत्मिश्रमृदि तत्क्षणजन्मकम्।
फलभान्विता शाखा भवतीति किमद्भुतम्।।
Vaapithamkarakonmisramrudi thathkshanjanmakam

Scanned with OKEN Scanner

phalabhaarnaanvithaasaakhaa bhavatheethi kimadbhutham

The seeds are watered with melted hailstones, then they will sprout immediately and after reaching the flowering state they will become heavy with fruits and flowers.

This is claimed to be one of recent techniques adopted for the production of seedlings of seeds which take weeks and months together for sprouting. Here the cold ice appears to give a shock treatment to the seeds and fastens the sprouting process.

श्लेष्मातकस्य बीजानि निष्कुलीकृत्य भावयेत् प्राज्ञः
अङ्कोलविज्जलाद्बिश्छायायां सप्तकृत्वैवम्।।

Sleshmaathakasya beejaani nishkuleekruthya bhavayeth praajna: angolavijjalaadbhiscchaayaayaam sapthakruthvaivam

The fruits of shleshmataka tree are decorticated carefully and treated with the pulp of angola fruits seven times repeatedly and each time drying under the shade. This process will be useful to fasten the sprouting within a day, say the learned people.

This can be a subject of detailed study so that more useful data will be available of the use of agriculturists and farmers.

माहिषगोमयघृष्टान्यस्य करीषे च तानि निक्षिप्य
करकाजलमृद्योगे न्युप्तान्यह्ना फलकराणि।।

Maahishagomayaghrushtaanyasya kareeshe cha thaani nikshipya karakaajalamrudyoge nyupthaanyahnaa phalakaraani

Then they should be kept under the heap of powdered buffallo dung on soil wetted with melted hailstones. Then they will sprout very quickly.

As explained earlier this procedure has been acknowledged by the modern science also. Some of the seeds are stored prior to sowing in the cow dung.

Scanned with OKEN Scanner

These lines show that the agricultural practices in ancient India have many things in similar to modern science.

ध्रुवमृदुमूलविशाखा गुरुभं श्रवणस्तथाश्विनी हस्तः।
उक्तानि दिव्यादृग्भिः पादपसंरोपणे भनि।।

Dhruvamrudumoolavisaakhaa gurubham sravanasthathaasvinee hastha: ukthaani divyaadrgbhi: paadapasamropane bhaani

Plants should be planted or sown when the moon is passing through favourable lunar mansion which are the three nakshthras i.e Utthara, Uttharashada and Uttharabhadra, known as the Dhruvaas or the three end stars. Anuraadha, Chithra, Margaseersha, Revathi, known as the mild asterisms and on the Mula, Visaakha, Pushya, Sravana, Asvini and Hastha, also

From ancient period, in India the stars are directly connected with the harvesting and also field preparation and sowing. Here, too same approach can be seen in the agriculture practices.

✡ ✡ ✡

Some verses from Kashyapa Vrukshaayurveda are also equally interesting. These lines show that the overall approach followed by the ancient Indians appears to be the same in planting the trees and following agricultural practices. Thus we can see in the Kashyapa Vrukashaayurveda similar lines:

दूर्वावीरणसंयुक्ता सानूपा मृदुमृत्तिकाः तत्र
वाप्यः शुभा वृक्षाः सुगन्धिफलशाखिनः

Doorvaaveeranasamyukthaa saanoopaa mrudumrutthikaa:

Scanned with OKEN Scanner

thathra vaapya: subhaa vrukshaa: suganddhiphalasaakhina:

Land on which Bermuda grass (Cynodn dactylon) and viran cuscus grass (Andopogaon muricatus) are flouroshing, that soi will be soft and mellow and should be planted with good trees o medicinal or economical importance.

अशोकचम्पकारिष्टपुन्नागश्च प्रियङ्गव:। शिरीषोदुम्बरा: श्रेष्टा:
पारिजातकमेव च एते वृक्षा: शुभान्जनेया:
प्रथमं तांश्च रोपयेत्। देवालये तथोद्याने गृहेषूपवनेषु च।।

Asokachampakaarishtapunnaagaascha priyangava:
sireeshodumbaraa: sreshtaa: paarijaakameva cha
ethe vrukshaa: subhaajneyaa: prathamam thaamscha ropayeth
devaalaye thathodyaane gruheshoopavaneshu cha

The above plants are mentioned in the Varahamihira's Bruha samhitha. Three more trees are incorporated by Kashyapa, to th list given by Varahamihira. Champka (Michelia Champaka) Udumbara the cluster fig tree and Parijaatha which is the nigh jasmine.

The cause of plant ailments and treatment for those problem of the plants were also given by Kashyapa.

शाखाविटपपत्रैश्च छायया विहितैश्च ये। येपि पर्णफलैर्हीना रूक्षा पत्रैश्च पाण्डुरै:।। शीतोष्णवर्षवाताद्यैर्मूलैर्व्यामिश्रितैरपि। शाखिनां तु भवेद्रोगो द्विपानां लेखनेन च।। चिकित्सितेषु कर्तव्या ये च भूय: पुनर्नवा: शोधयेत् प्रथमं शस्त्रै: प्रलेपं दापयेत् तत:।। कर्दमेन विडङ्गैश घृतमिश्रैश्च लेपयेत्। क्षीरतोयेन सेक: स्याद् रोहणं सर्वशाखिनाम्।।

Sakhaavitapapatharaischa cchayayaa vihithaischa ye
yeapi parnaphalairheenaa rookshaa pathraischa paandurai:
seethoshnavarshavaathaadyairmoolairvyaamisrithairapi
saakhinaam thu bhavedrogo dwipaanaam lekhanena cha
chikitsitheshu karthavyaa ye cha bhooya: punarnavaa:

Scanned with OKEN Scanner

sodhayeth prathamam sasthrai: pralepam daapayeth thatha:
kardamena vidangaischa ghruthamisraischa lepayeth
ksheerathoyena seka: syad rohanam sarvasaakhinaam

Due to the shade and non availability of the sunlight, the plants can become leafless and fruitless. Similarly their leaves can also be of blurd coloured. Extreme climatic conditions of cold, heat, rain and twinning roots and branches can also in the plant. Distrubances from the animals can also can be diseases. It is the duty to treat those plants for rejuvenating them. First of all clean the plant, scientifically, with dilute milk on the branches and then given a coating of Dolichos biflorus plant dicoction with green gram and black gram by spraying the solution. These lines are in agreement with the modern observations.

Scanned with OKEN Scanner